



A+ LAB SERIES V5

Lab 16: Troubleshooting Tools in Windows

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Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	5.1: Given a scenario, troubleshoot motherboards, RAM, CPUs, and power.
CompTIA A+ (220-1202) Exam Objectives	1.4: Given a scenario, use Microsoft Windows operating system features and tools. 3.1: Given a scenario, troubleshoot common Windows OS issues.

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Introduction

For optimal operation, computers and networks require all components to function as expected, providing a seamless experience for users and technicians alike. However, when issues arise, a structured troubleshooting methodology is essential for resolving problems effectively.

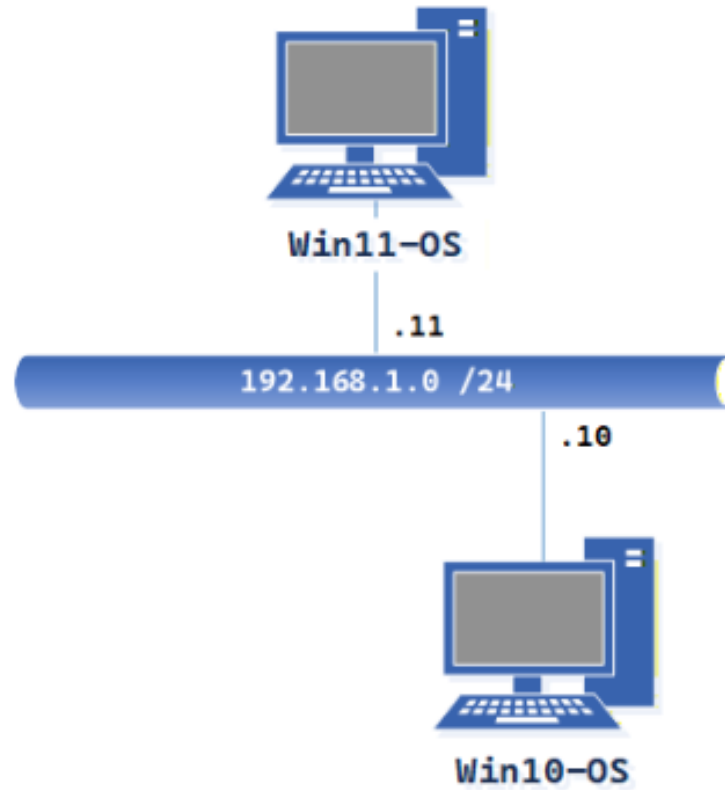
Troubleshooting involves systematically identifying and resolving issues, where pinpointing the root cause is often the most challenging step. By following a logical approach, technicians can streamline this process, leading to quicker, more efficient resolutions.

Objective

In this lab, you will explore techniques to troubleshoot and automate administration in Windows 10 and 11, using available tools and methods now that the Action Center has been replaced with the Notification Center, to reinforce your understanding of troubleshooting tools in Windows.

- Use Windows 11's new Action Center to troubleshoot hardware.
- Use Task Manager to monitor applications, processes, and services. Create a dump file. Use Administrative Tools in the Control Panel to see options.
- Check System Reliability(Reliability Monitor) through Security and Maintenance.
- Examine the Event Viewer and save events.
- Use System Configuration Utility to analyze the performance, boot options, services, startup, and tools. Change boot options and compare logs.
- Analyze system files using the System File Checker in the Command Prompt.
- Use Windows Recovery Environment to see options or use DISM.
- Use the Steps Recorder to record events.

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

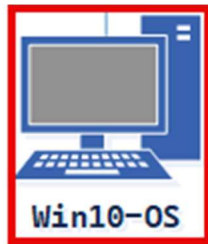
Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!
Win11-OS	192.168.1.11 / 24	sysadmin	NDGLabpass123!

1 Windows Troubleshooting Tools

In this section, you will explore fundamental Windows troubleshooting methods. Windows 10 and 11 include a variety of tools designed to maintain system health and performance. These tools can be easily accessed through the Control Panel and the Search feature in both versions.

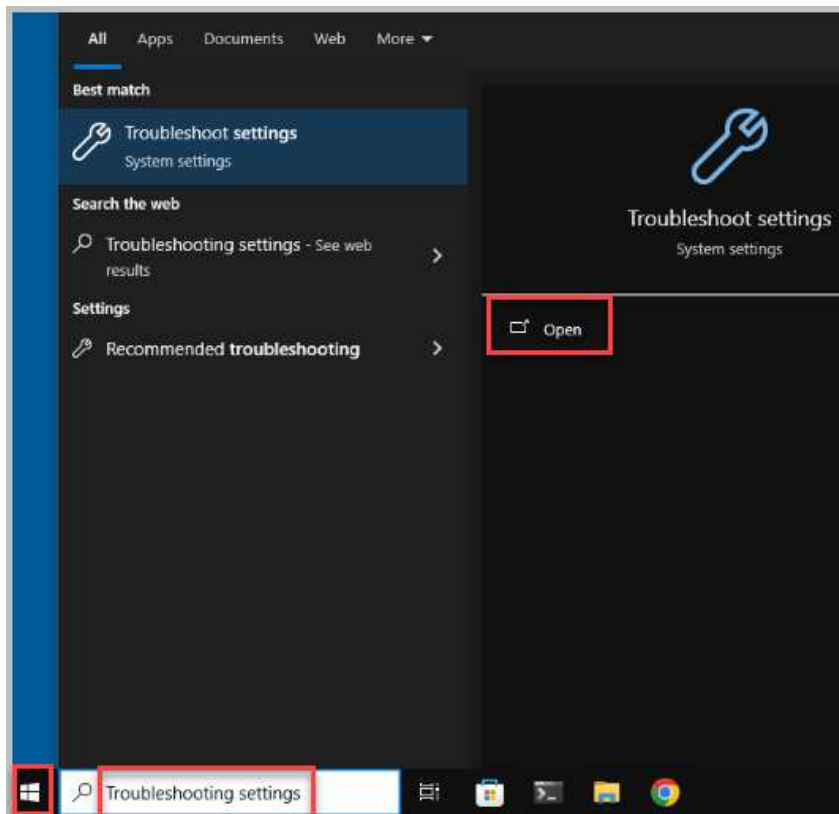
1.1 Action Center and Troubleshooters

1. Open a console connection to the **Win10-OS**.



To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

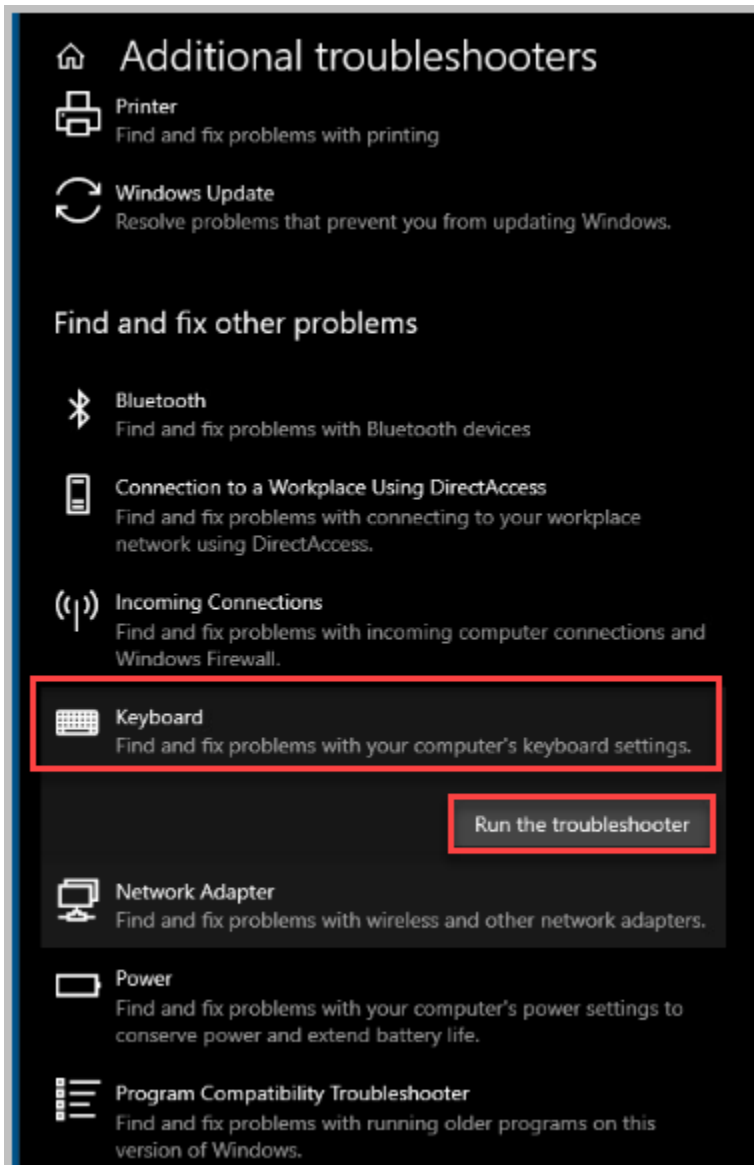
2. Click on the **Start** button, followed by typing **Troubleshoot settings** in the search field. Select **Open**.



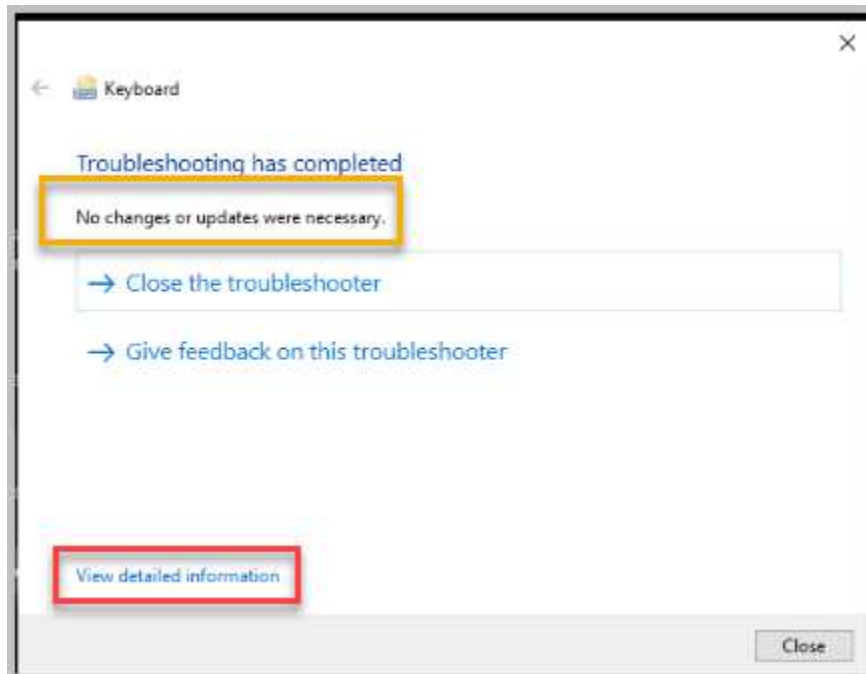
3. Select **Additional troubleshooters**.



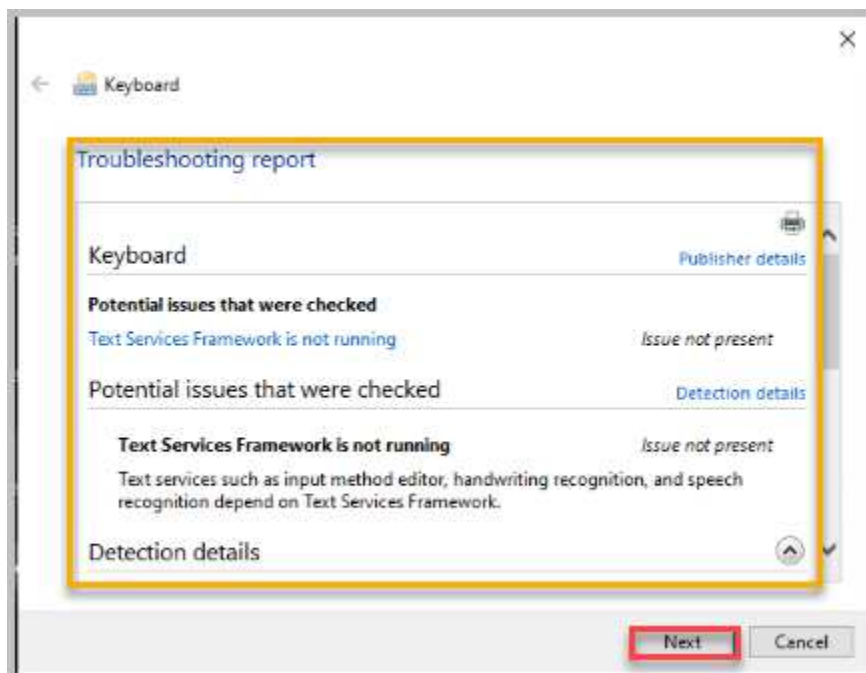
4. In Windows 10, Additional Troubleshooters provides a collection of built-in tools that help resolve common issues related to connectivity, hardware, and system performance. Review the list of *Additional troubleshooters* for both *Get up and running* and *Find and Fix other problems*. Select **Keyboard** and select **Run the troubleshooter**.



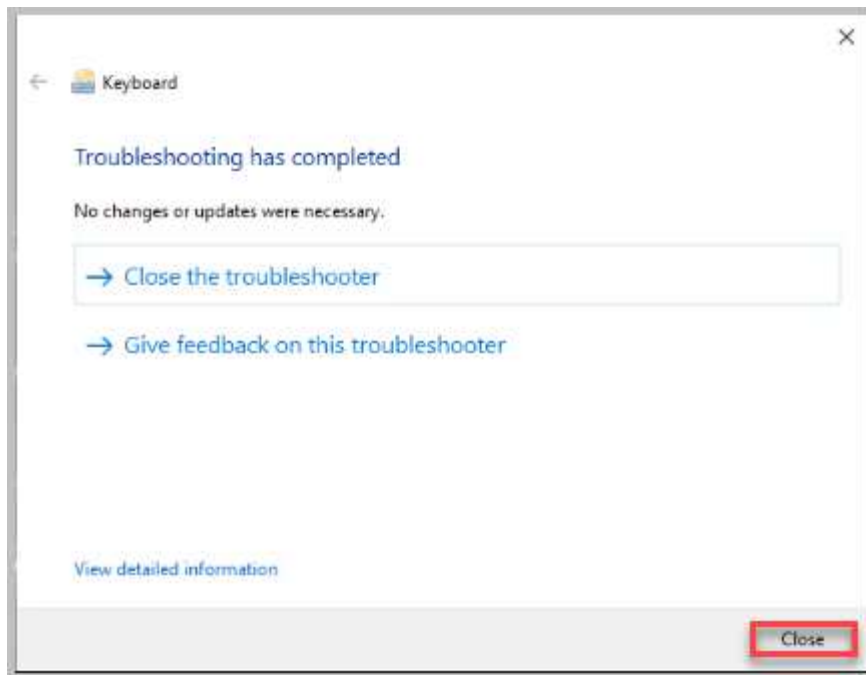
5. Notice the *Troubleshooting* has completed, and no changes or updates were necessary. Select **View detailed information**.



6. Expand each item within the *Troubleshooting report*, and review the items checked with the *Keyboard troubleshooter*. Select **Next**.



7. Select **Close** for the Keyboard troubleshooter.



Challenge Task:

As a challenge, try to recreate this task in a different version of *Windows OS* that is made available in this lab environment. This will help show similarities and differences between the versions.

8. Close all open windows and leave the *Windows 10* console open to continue with the next task.

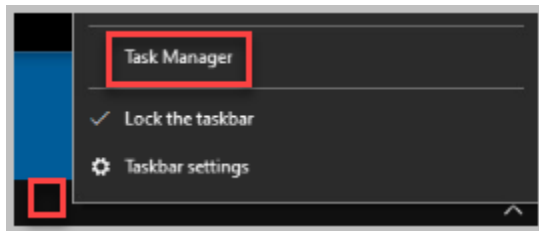
1.2 Managing Processes with Task Manager

Sometimes it is necessary to escalate an issue to a technician with more advanced skills, but they will require information or additional documentation about a problem. *Task Manager* is a tool for creating documentation to help resolve such problems.

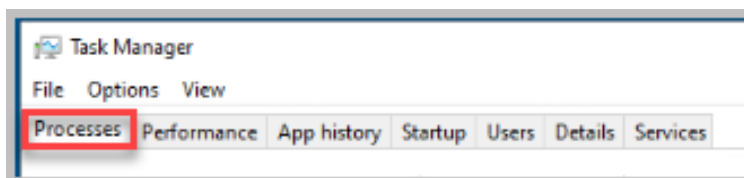
1. Select **File Explorer** on the taskbar.



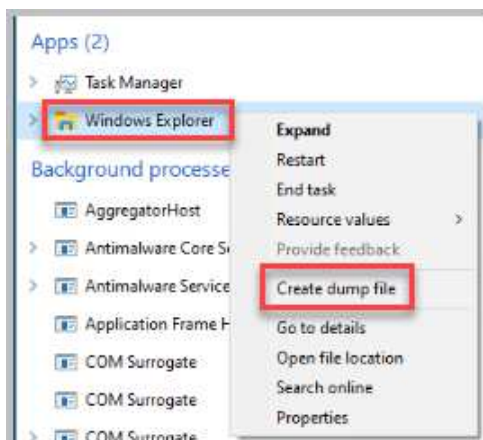
2. Right-click on the taskbar and select **Task Manager**.



3. This launches the *Windows Task Manager*. This tool monitors Processes, Performance, App history, Startup, Users, Details, and Services running on the system. *Task Manager* can be used to start and stop programs and processes. It will also show you information about your computer and network performance. Ensure that you are on the **Processes** tab.



4. Right-click on **Windows Explorer** and click **Create Dump File**.

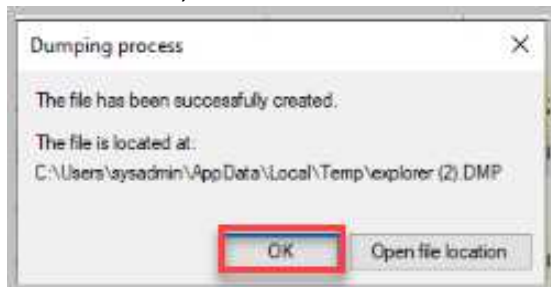


Dump files are useful when used with *Windows' Debugger* tool. A dump file is a snapshot of an app at the point in time the dump was taken. It shows what process was executing and what modules were loaded. It creates a file that contains information that a technician or programmer might find useful in finding problems with the application. If applications are freezing up, this is a good method of troubleshooting the issue.

5. Notice the pop-up window. Wait 1-2 minutes for the dump file to be created.



6. Once created, click **OK**.



The dump file can now be used to send to the correct person to escalate support. The file is a record of everything that was in the computer's memory when a crash occurred, and it's meant to help diagnose what went wrong.

7. Close the **Task Manager** and **File Explorer** windows.



Challenge Task:

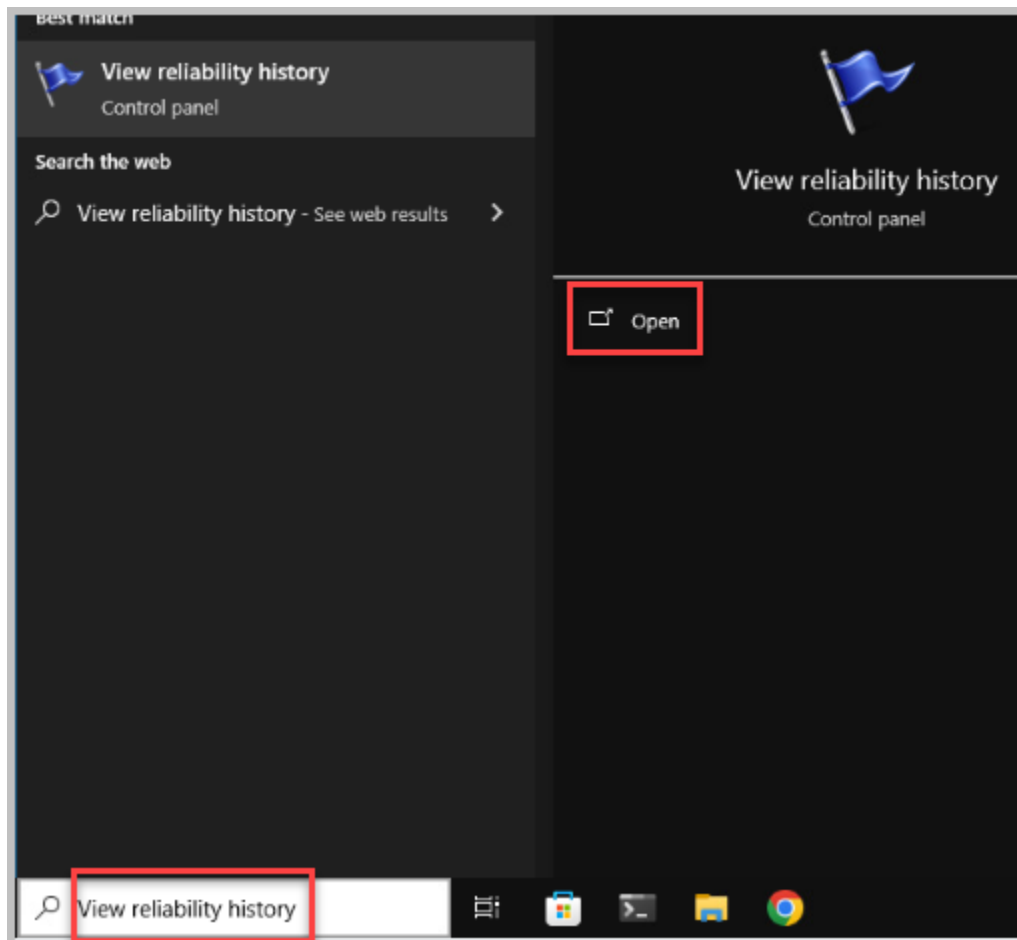
As a challenge, try to recreate this task in a different version of *Windows OS* that is made available in this lab environment. This will help show similarities and differences between the versions.

8. Leave the *Windows 10* window open to continue with the next task.

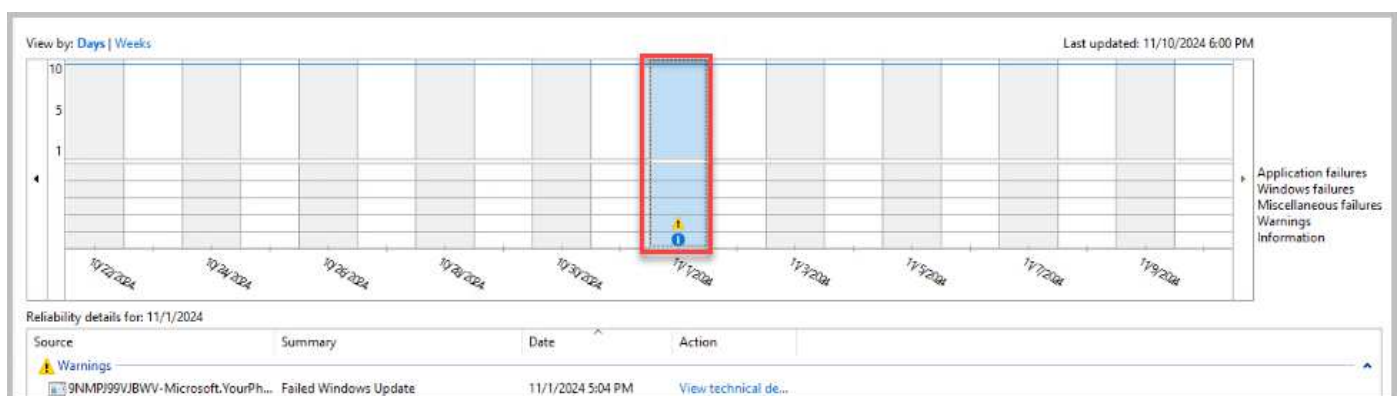
1.3 Check System Reliability

If a computer exhibits strange behavior over a period, you can use the *Reliability Monitor* to view your computer's reliability and problem history.

1. Type **View reliability history** in the search box on the taskbar and select **Open**.



2. Notice a graph appears with data. By default, it should be selected for your current day. You can select any day by clicking on the chart above the printed dates.



In the graph, problems are shown as red crosses, warnings are shown as yellow exclamation marks, and information is shown in blue 'i' icons.

3. Notice in this example, a problem is listed where the *Notepad* application stopped responding. You may not see this problem, as it was listed on 11/1/2024. You may try creating problems in the Windows 10 operating system to view the associated problems on your graph. At the bottom of the page, there are links to save a history, view reports, and check for a solution.



Challenge Task:

As a challenge, try to recreate this task in a different version of Windows OS that is made available in this lab environment. This will help show similarities and differences between the versions.

4. Close the **Reliability Monitor** window.
5. Leave *Win10-OS console* open to continue with the next task.

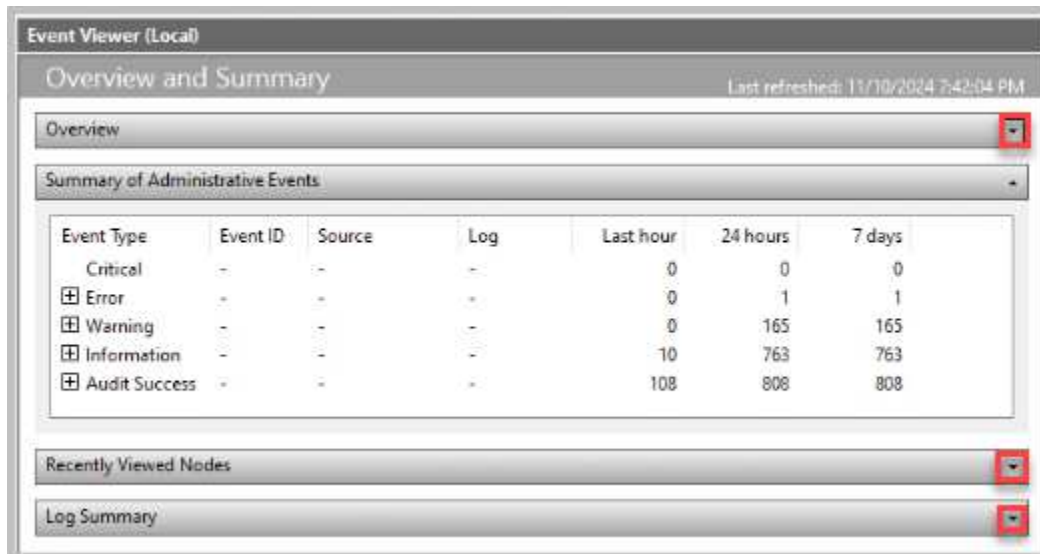
1.4 Examining Event Viewer

Event Viewer is a tool that tracks information about momentous events on your computer. *Event Viewer* can be helpful when troubleshooting issues with *Windows* and other programs, hardware, and security events.

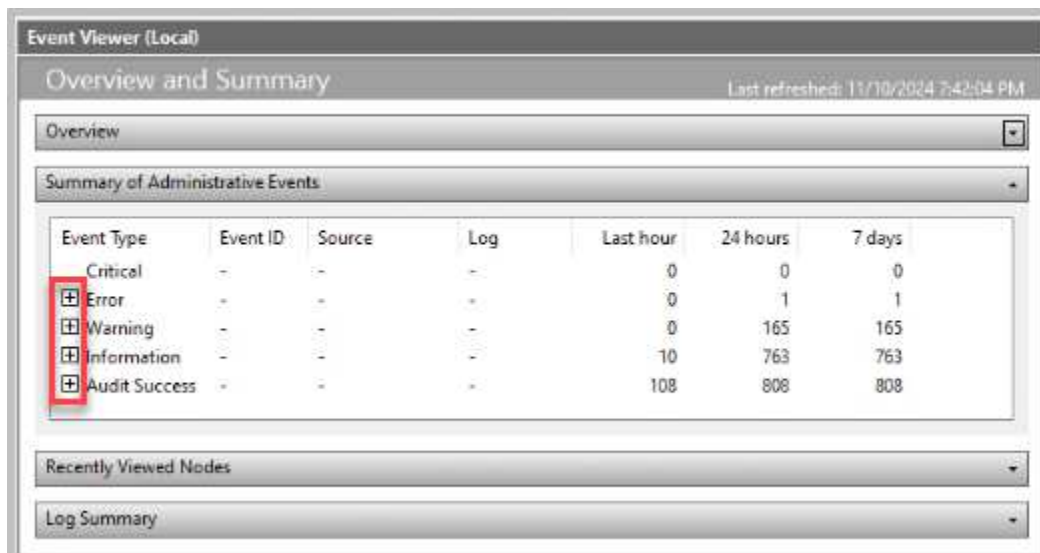
1. Right-click the Start button and select **Event Viewer**.



- Notice the *Event Viewer* appears, and by default, it shows the *Overview and Summary* page. This page displays all events from all *Windows* logs on the system. The total number of events for each type that have occurred is displayed, along with the number of events of each type that have occurred over the last seven days, the last 24 hours, and the last hour. In the middle pane, click the **small up arrows** at the end of the section headers to minimize all but the **Summary of Administrative Events** section.

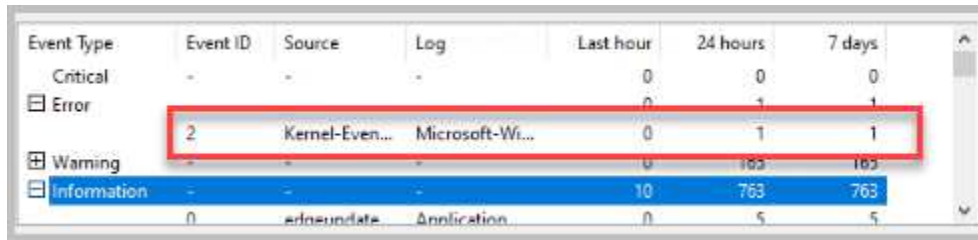


- Viewing the *Summary of Administrative Events* section allows you to view the diverse types of events that may have occurred on this system. Click the + next to each type of event to view the related events that have occurred on this system.



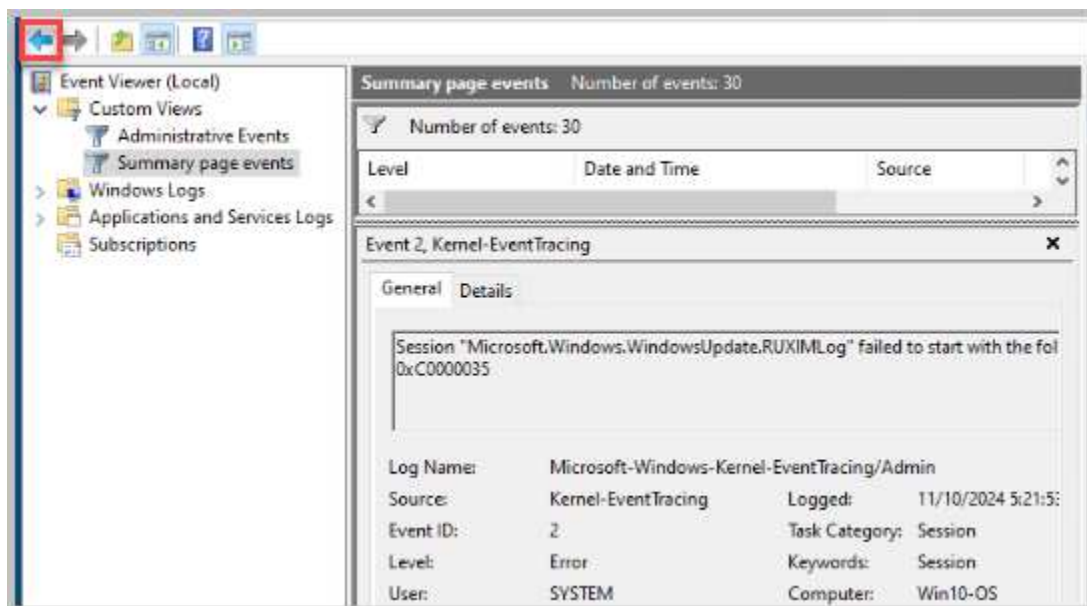
The different event types: *Critical*, *Error*, *Warning*, *Information*, and *Audit Success*.

4. Select any specific event and then double-click it to display detailed information on the event. Due to the date change, your events may be different from the example used. In this example, an *Error* event with an ID of 2 is selected.

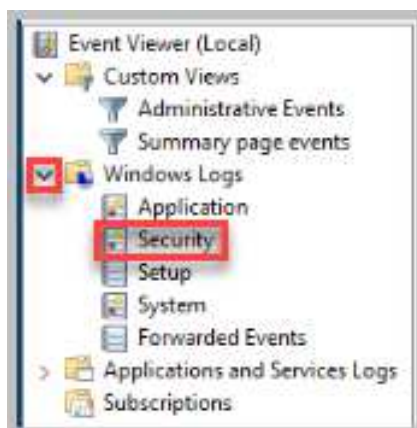


Event Type	Event ID	Source	Log	Last hour	24 hours	7 days
Critical	-	-	-	0	0	0
Error	2	Kernel-Even...	Microsoft-Wi...	0	1	1
Warning	-	-	-	0	165	165
Information	-	-	-	10	763	763
	0	edgeupdate	Application	0	5	5

5. Notice more details emerge for the specific event. In this example, *Windows Update failed to start*. Click the **back arrow** when finished analyzing the specific event.



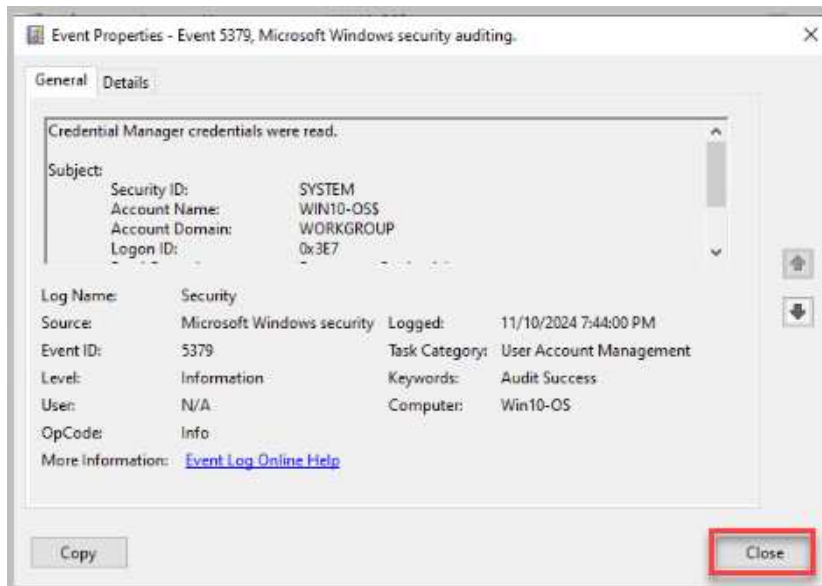
6. In the left pane, expand **Windows Logs** and click on **Security**.



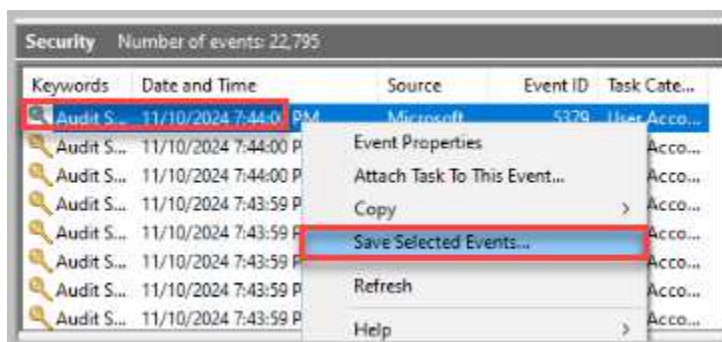
7. In the middle pane, double-click any security-specific event to display detailed information on the corresponding event.



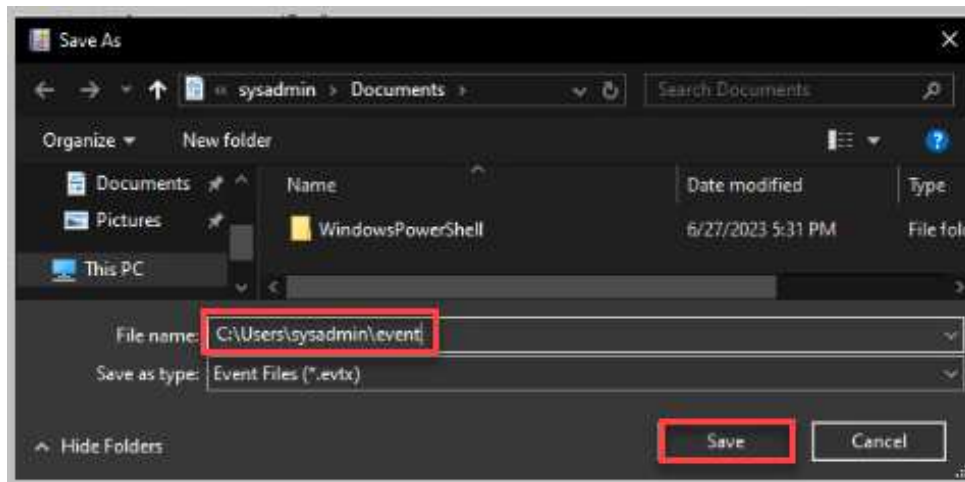
8. Review the available information and click **Close**.



9. Right-click the same event you just opened and select **Save Selected Events**.



10. In the *Save As* window, type `C:\Users\sysadmin\event` in the *File name* text field and click **Save**.



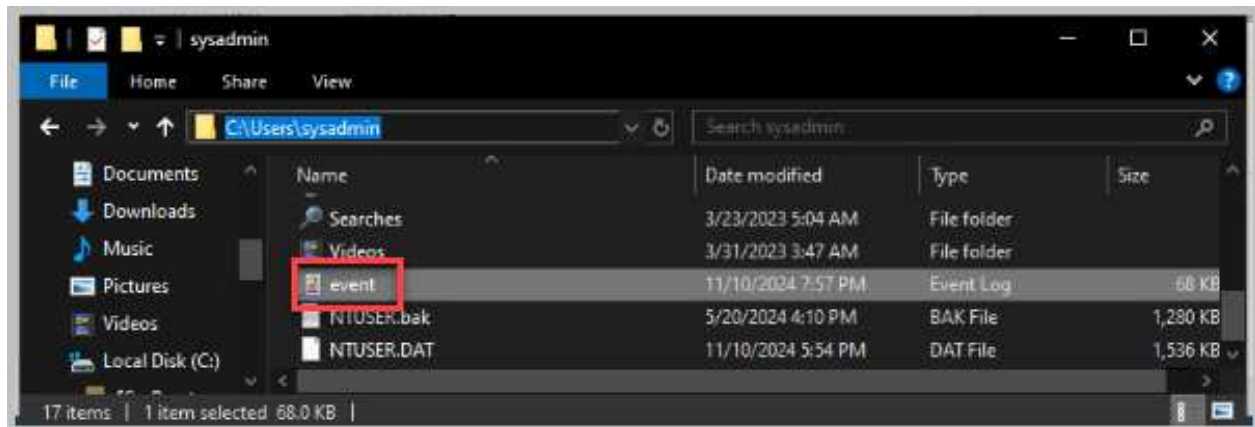
11. When prompted about display information, leave the defaults and click **OK**.



12. Open **File Explorer**.



13. In the *Windows Explorer* window, navigate to **C:\Users\sysadmin** and locate the **event** file. Double-click on the **event** file entry.



14. Notice that the event file launches the *Event Viewer* application to view the detailed information saved. Close the window when you are finished reviewing the information.



Challenge Task:

As a challenge, try to recreate this task in a different version of *Windows OS* that is made available in this lab environment. This will help show similarities and differences between the versions.

15. Close all remaining windows in *Win10-OS*.

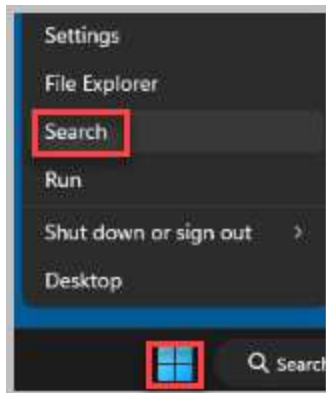
1.5 Using the System Configuration Utility

The *System Configuration (MSConfig)* utility is a built-in program in the *Windows* operating system. Using *MSConfig*, you can analyze a computer's performance by comparing it when certain applications are running versus when they are disabled. It allows access to modify the startup settings that *Windows* applies. *MSConfig* can be useful for solving startup problems. It is also very useful for troubleshooting boot issues. *Msconfig* creates a boot log file, *ntbtlog.txt*, at each boot. The log records all information from each boot process, including drivers loaded during normal boot and those loaded during *Safe Mode* boot. The log can be read with any text editor.

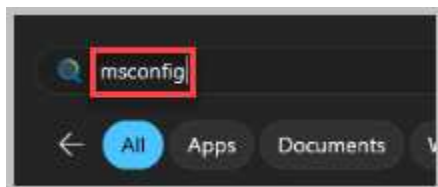
1. Open a console connection to the **Win11-OS** system.



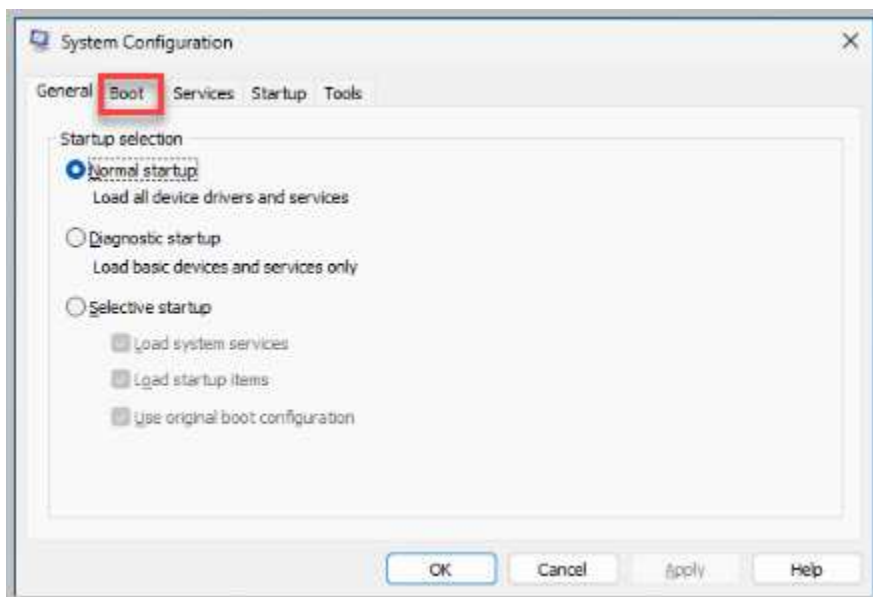
2. Right-click the **Start** button and select **Search**.



3. Type `msconfig` in the search field, and press the **Enter** key.



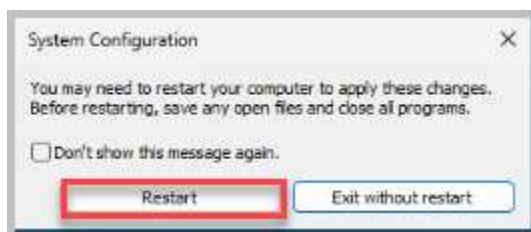
4. In the *System Configuration* window, click on the **Boot** tab.



5. Check the checkbox next to **Safe boot** and ensure that the **Minimal** radio button is selected. Make sure that the **Boot log** option is also checked. Click **OK**.



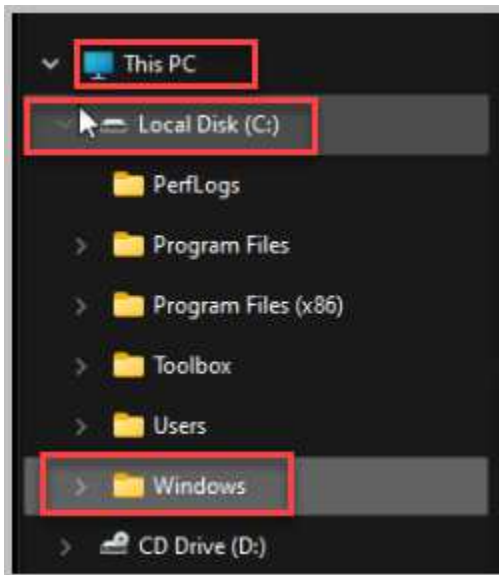
6. In the *System Configuration* pop-up, select **Restart**.



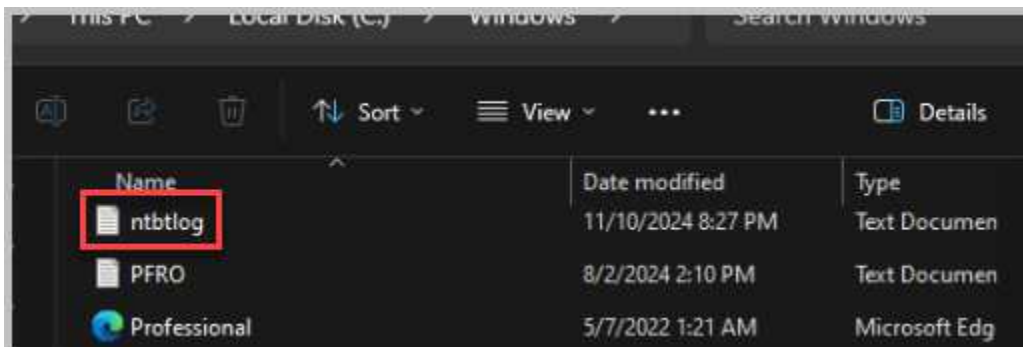
7. Once the system finishes rebooting, *Win11-OS* will automatically boot in with the account **sysadmin**.
8. Notice the overall change in desktop appearance when in *Safe Mode*. Select **File Explorer** on the taskbar.



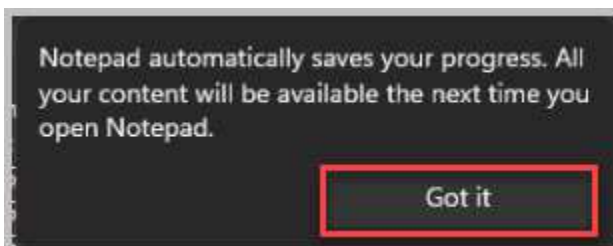
9. Expand **This PC**, **Local Disk (C:)**, and select **Windows**.



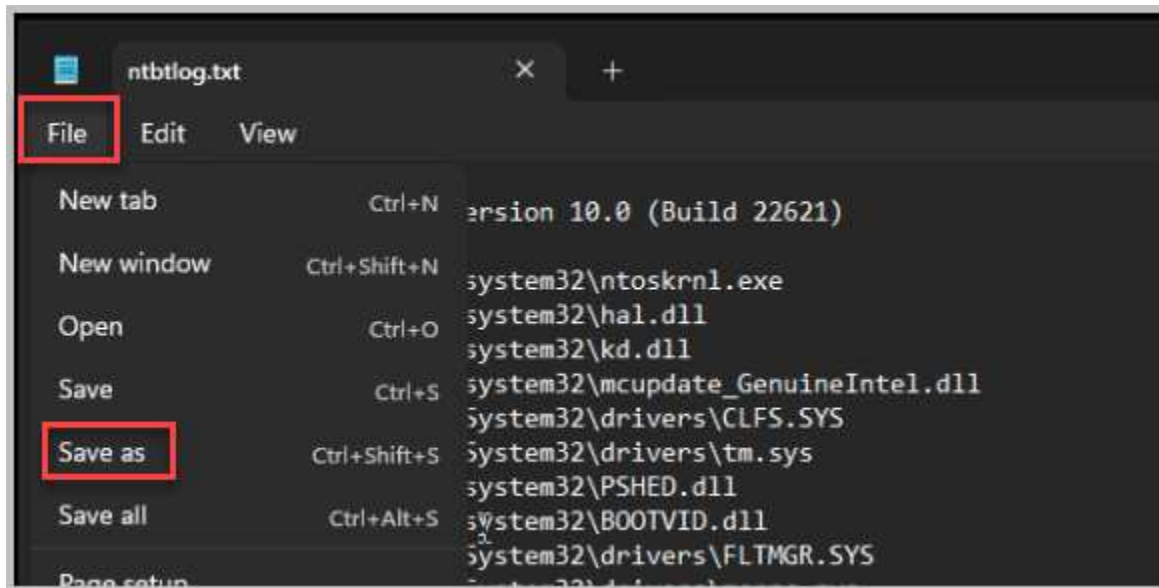
10. In the right pane, scroll the list available and double-click **ntbtlog**.



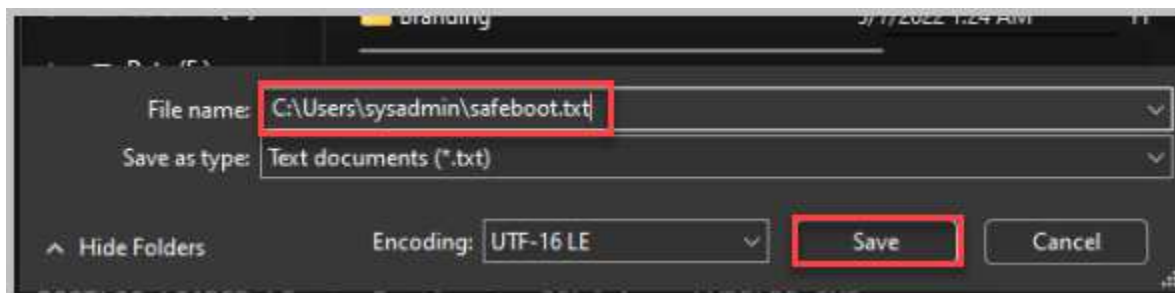
11. Select **Got it** to acknowledge that *Notepad automatically saves your progress*.



12. In the *Notepad* window, notice the log contents of the safe boot. Click **File** and select **Save As**.

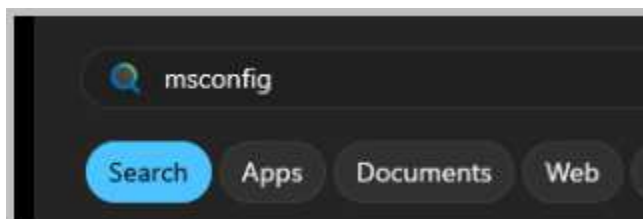


13. In the *Save As* window, type **C:\Users\sysadmin\safeboot.txt** in the *File name* text field and click **Save**.

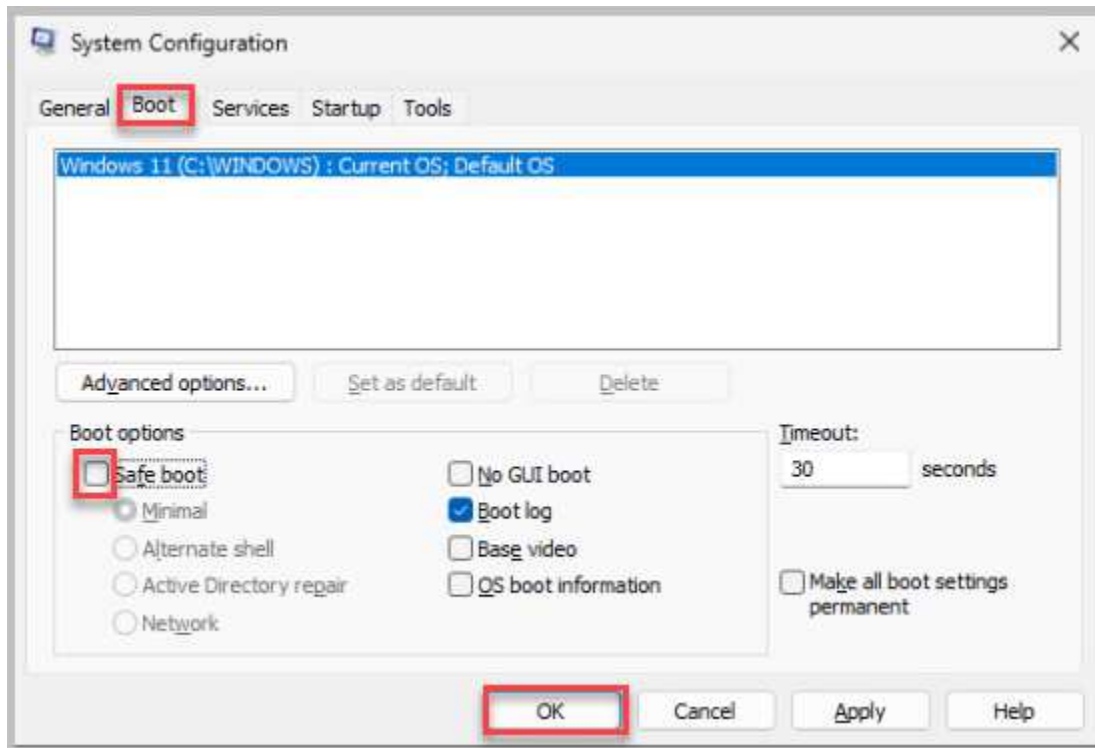


14. Once saved, close the **Notepad** application and **Windows File Explorer** window.

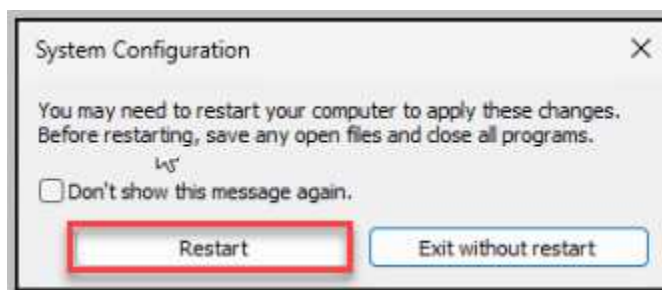
15. Click the **Start** button, type **msconfig** in the search field, and press the **Enter** key.



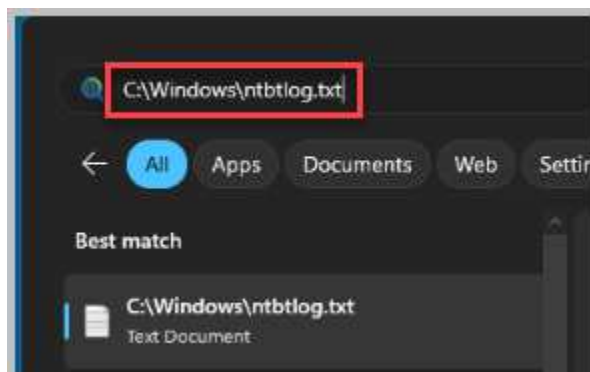
16. In the *System Configuration* window, click on the **Boot** tab. **Uncheck** the checkbox for **Safe boot** and click **OK**.



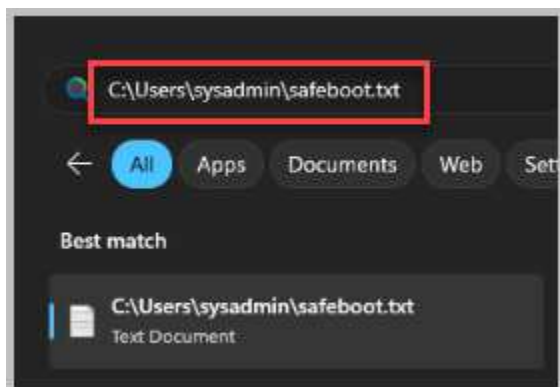
17. Select **Restart** to reboot the *Win11-OS* system.



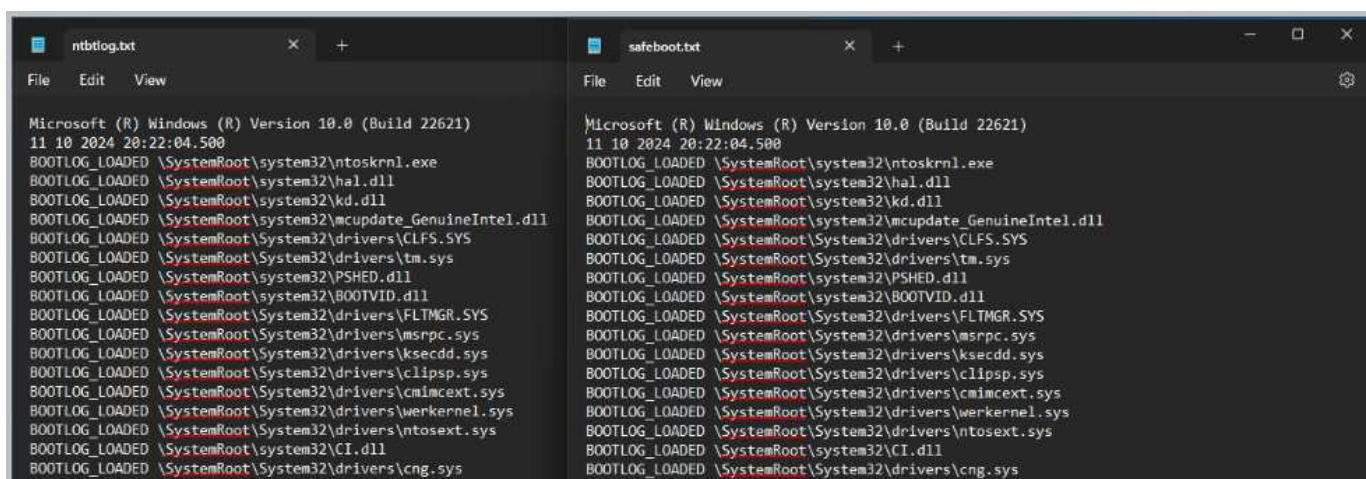
18. Once logged in, click the **Start** button, type `C:\Windows\ntbtlog.txt` in the search field, followed by pressing the **Enter** key.



19. Open another file. Click the **Start** button, type `C:\Users\sysadmin\safeboot.txt` in the search field, and press the **Enter** key.



20. Arrange the two *Notepad* windows side-by-side for comparison. This is how you can troubleshoot and identify differences between the drivers loaded during the two separate system bootups. These logs are also used to share with vendor tech support.



21. Close both **Notepad** windows.



Challenge Task:

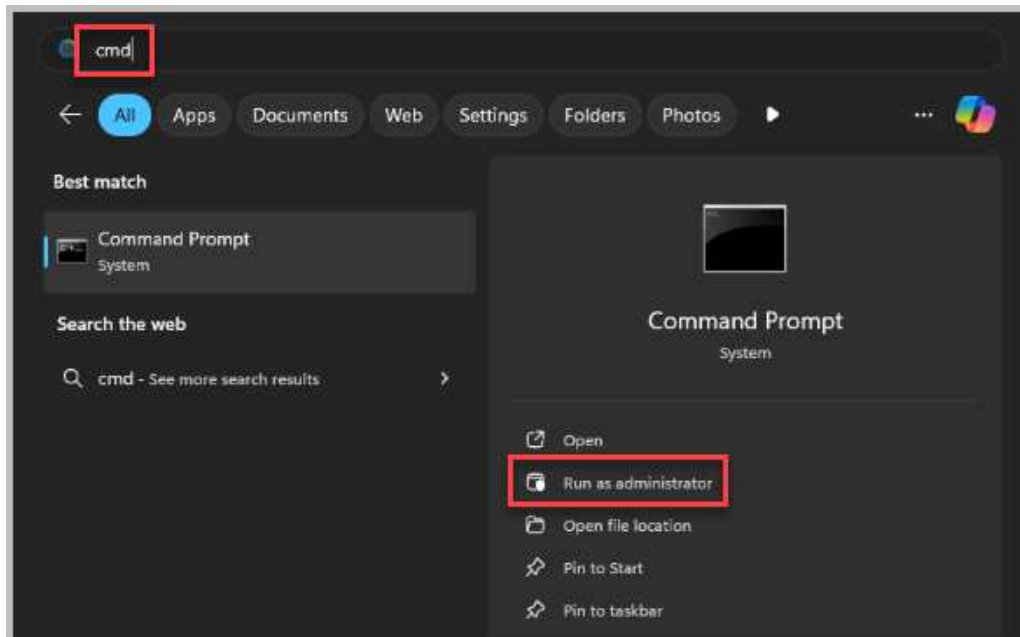
As a challenge, try to recreate this task in a different version of *Windows OS* that is made available in this lab environment. This will help show similarities and differences between the versions.

22. Leave the *Win11-OS* window open to continue with the next task.

1.6 Analyzing System Files

Using *SFC* in *Windows* scans your drive for corrupt or missing system files and replaces any it finds with working versions, which can help fix many other odd problems that may have been caused by corruption.

1. Click the **Start** button and type **cmd** in the search field. Right-click on **cmd.exe** from the search results and select **Run as administrator**.

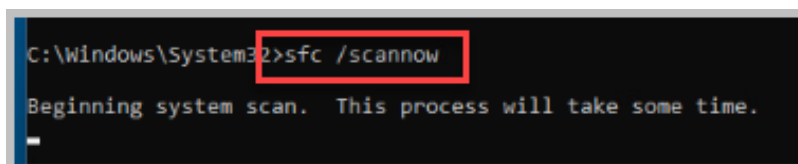


2. When prompted with User Account Control, click **Yes** to continue.



3. Type the command below, followed by pressing the **Enter** key.

```
sfc /scannow
```



**Please
Note**

The *SFC* scanning process can take 5-7 minutes to complete.

- Once completed, the *SFC* utility reported back with corrupt files found and successfully repaired them. If you were needing to research the exact issue, they are available at `C:\Windows\Logs\CBS\CBS.log`.

```
C:\Windows\System32>sfc /scannow

Beginning system scan. This process will take some time.

Beginning verification phase of system scan.
Verification 100% complete.

Windows Resource Protection found corrupt files and successfully repaired them.
For online repairs, details are included in the CBS log file located at
windir\Logs\CBS\CBS.log. For example C:\Windows\Logs\CBS\CBS.log. For offline
repairs, details are included in the log file provided by the /OFFLOGFILE flag.
```

- Close the **command prompt** window.

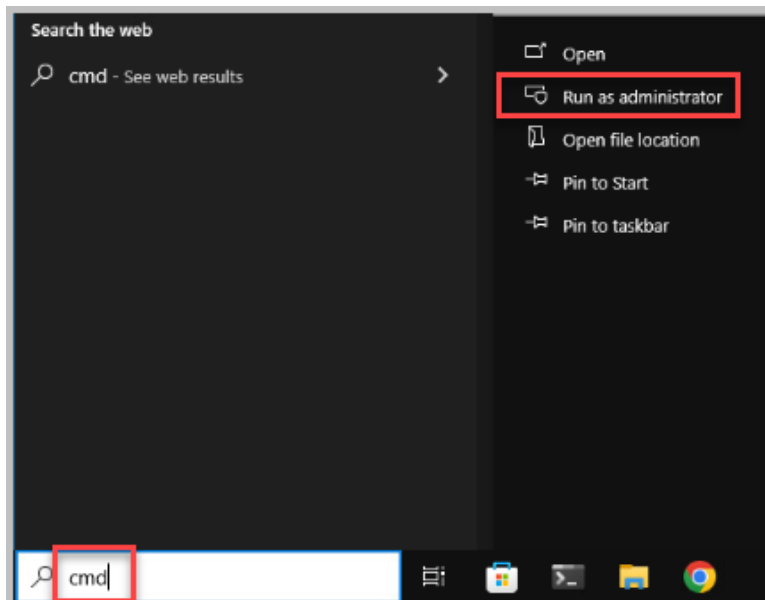
1.7 Using the Deployment Image Servicing and Management (DISM) tool

Deployment Image Servicing and Management (DISM) is a command-line program for *Windows* that repairs and prepares *Windows* images, including the *Windows Recovery Environment*, *Windows Setup*, and *Windows PE*, via the command line or *PowerShell*. It is often used when the *System File Checker* utility can find corrupt files but cannot restore missing system files. Run *DISM* from an elevated command prompt.

- Access the **Win10-OS** console.



- Click the **Search** button, type `cmd`. Right-click the Command Prompt listing and select **Run as administrator**.

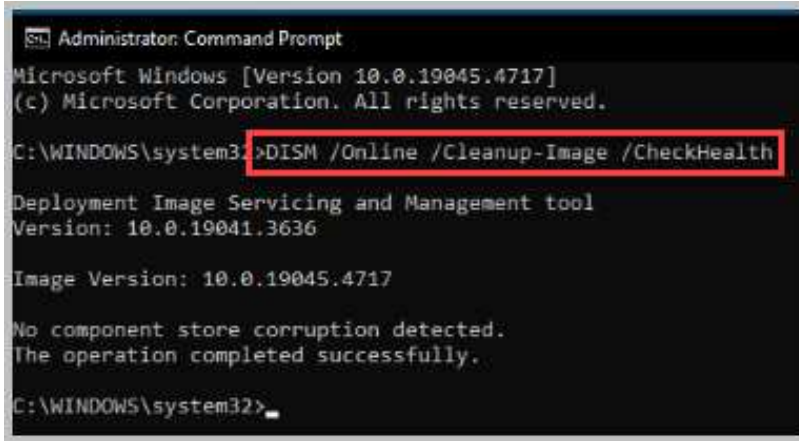


- When prompted, click **Yes** to allow the UAC to make changes.



4. In the command prompt window, type the command below, followed by pressing the **Enter** key. Using the */CheckHealth* switch quickly scans the local image for corruption and detects any issues, but does not run repair operations. This will take a few seconds to complete.

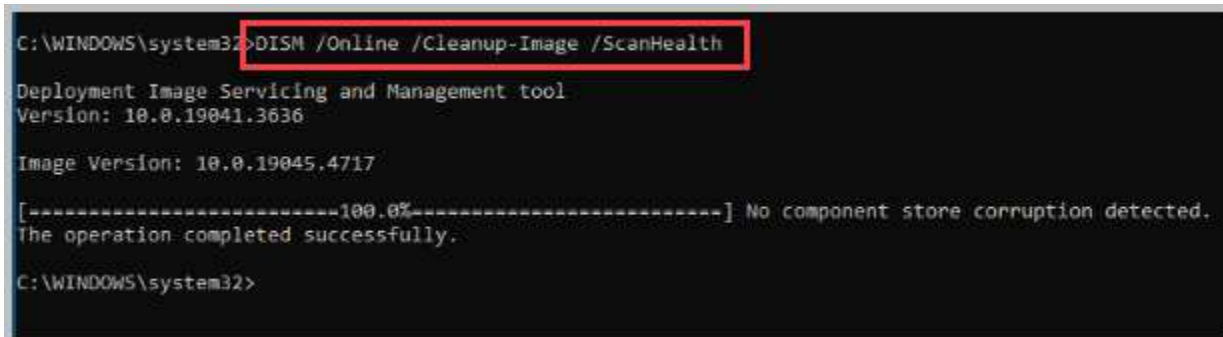
```
DISM /Online /Cleanup-Image /CheckHealth
```



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.4717]
(c) Microsoft Corporation. All rights reserved.
C:\WINDOWS\system32>DISM /Online /Cleanup-Image /CheckHealth
Deployment Image Servicing and Management tool
Version: 10.0.19041.3636
Image Version: 10.0.19045.4717
No component store corruption detected.
The operation completed successfully.
C:\WINDOWS\system32>
```

5. In the command prompt window, enter the command below. Using the */ScanHealth* switch initiates a more advanced scan of the local image but does not correct any issues. This will take approximately 4-6 minutes to complete.

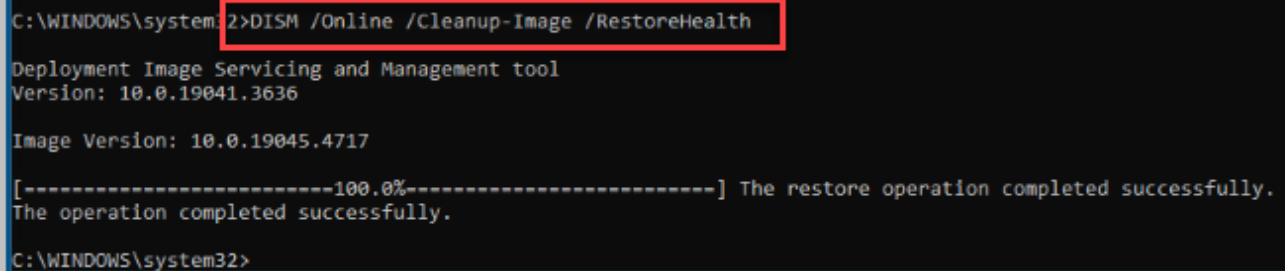
```
DISM /Online /Cleanup-Image /ScanHealth
```



```
C:\WINDOWS\system32>DISM /Online /Cleanup-Image /ScanHealth
Deployment Image Servicing and Management tool
Version: 10.0.19041.3636
Image Version: 10.0.19045.4717
[-----100.0%-----] No component store corruption detected.
The operation completed successfully.
C:\WINDOWS\system32>
```

6. In the command prompt window, enter the command below. Using the */RestoreHealth* switch scans for component store corruption and attempts to repair the *Windows 10* image. This will take several minutes to complete. Be patient and wait until it completes.

```
DISM /Online /Cleanup-Image /RestoreHealth
```



```
C:\WINDOWS\system32>DISM /Online /Cleanup-Image /RestoreHealth
Deployment Image Servicing and Management tool
Version: 10.0.19041.3636

Image Version: 10.0.19045.4717

[-----100.0%-----] The restore operation completed successfully.
The operation completed successfully.

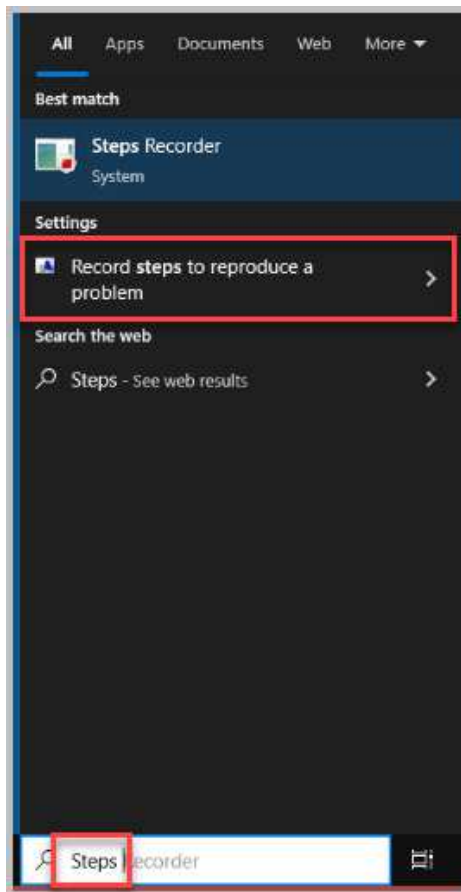
C:\WINDOWS\system32>
```

7. After the process is complete, close the command prompt window. Leave the Win10-OS system open for the next activity.

1.8 Using Documentation Tools

In addition to the many logs recorded by Windows, such as those in Event Viewer, Performance Monitor, and others, there is also the Problem Steps Recorder (known as Steps Recorder in Windows 10 and renamed in Windows 11) to aid in troubleshooting. It is a program that records the exact steps taken when the problem occurred to help troubleshoot a problem on your device. It can also be useful for remote support, as it allows a user to show you exactly what type of error message is received and the events that triggered it.

1. Click the **Start** button, type **Steps**, and click on the **Record steps to reproduce a problem** search result entry.



2. In the *Problem Steps Recorder*, click on the **Start Record** icon.



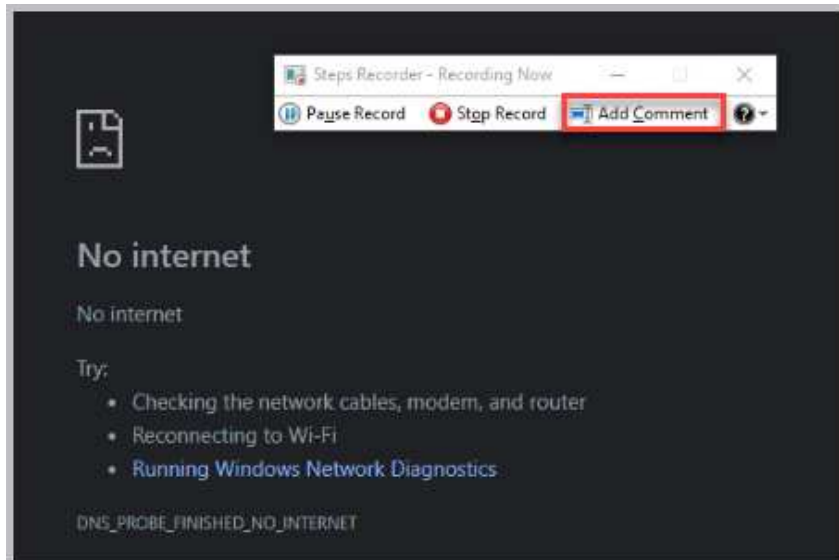
3. When recording, you normally go through the process of diagnosing a problem. You can pause and resume the recording at any time. To simulate a problem, open **Chrome** and record the steps you take to receive the error when trying to connect to the Internet. Notice, there is no active Internet connection in this environment.



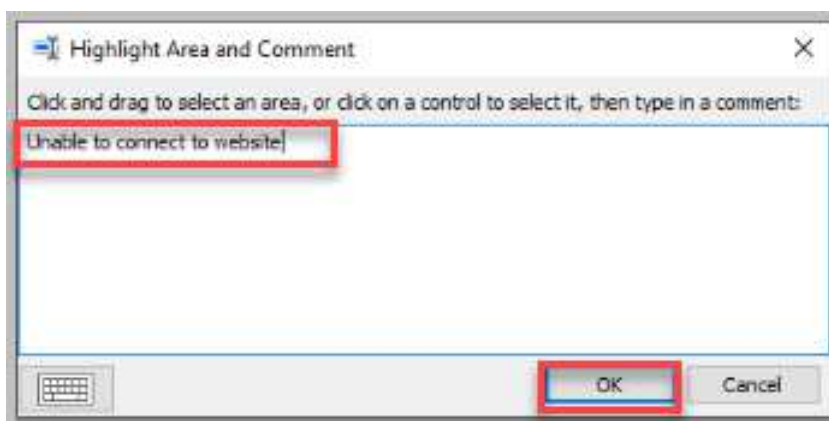
4. In *Firefox*, type `www.netdevgroup.com` in the address field and press the **Enter** key.



5. Wait until *Firefox* responds with a message. On the *Problem Steps Recorder* window, click on the **Add Comment** button.



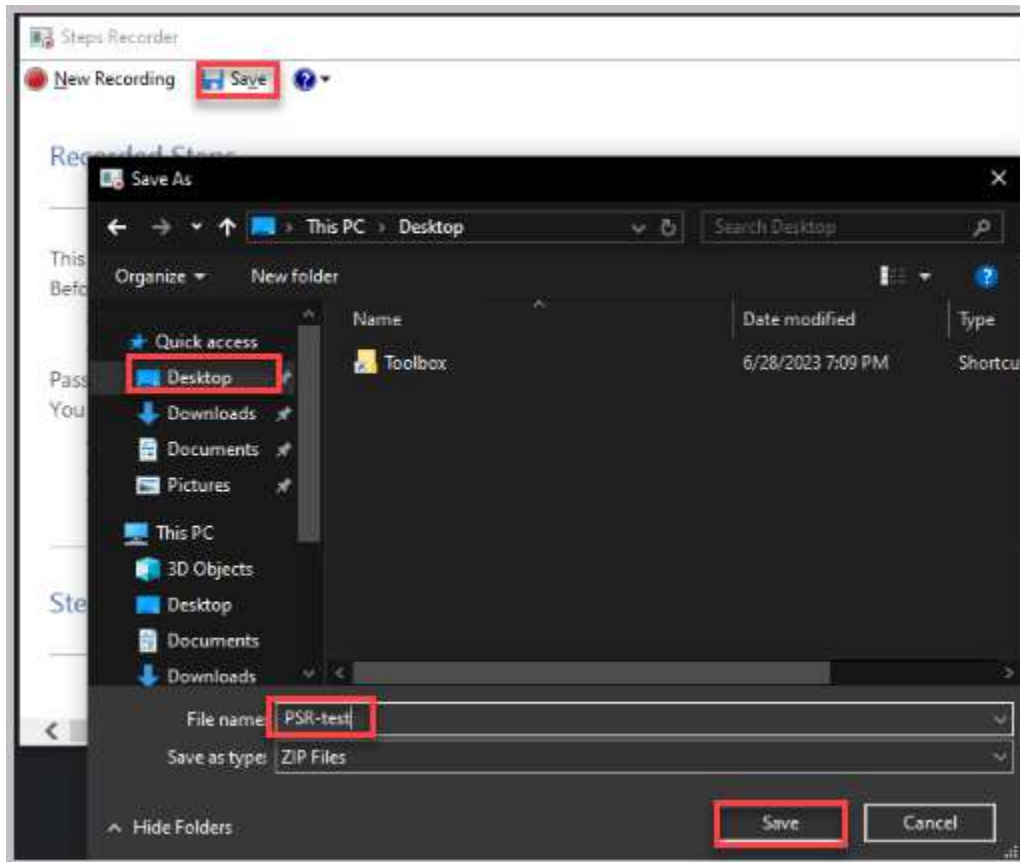
6. Use your mouse to select the part of the screen with the error message. Once the selection is made, type "Unable to connect to website" in the *Highlight Problem and Comment* window. Click **OK**.



7. When you are done, click **Stop Record**.



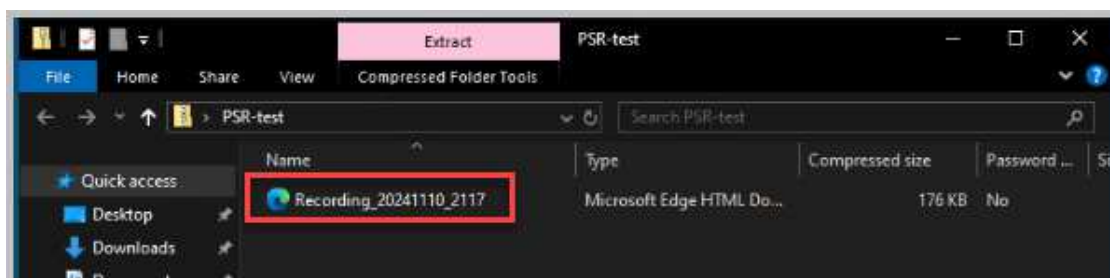
8. In the *Steps Recorder* window, select **Save**. Notice that a *Save as* window appears. Choose **Desktop** as the save to location and type **PSR-test** in the *File name* text field. Click **Save** to save the recording.



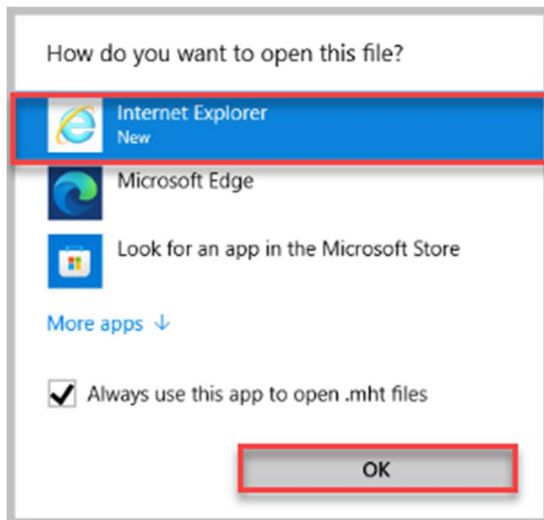
9. Close the open *Chrome Window*. Double-click the **PSR-test.zip** file on the *Desktop*.



10. Inside the zip file, double-click on the **Recording_*** file to view its contents.



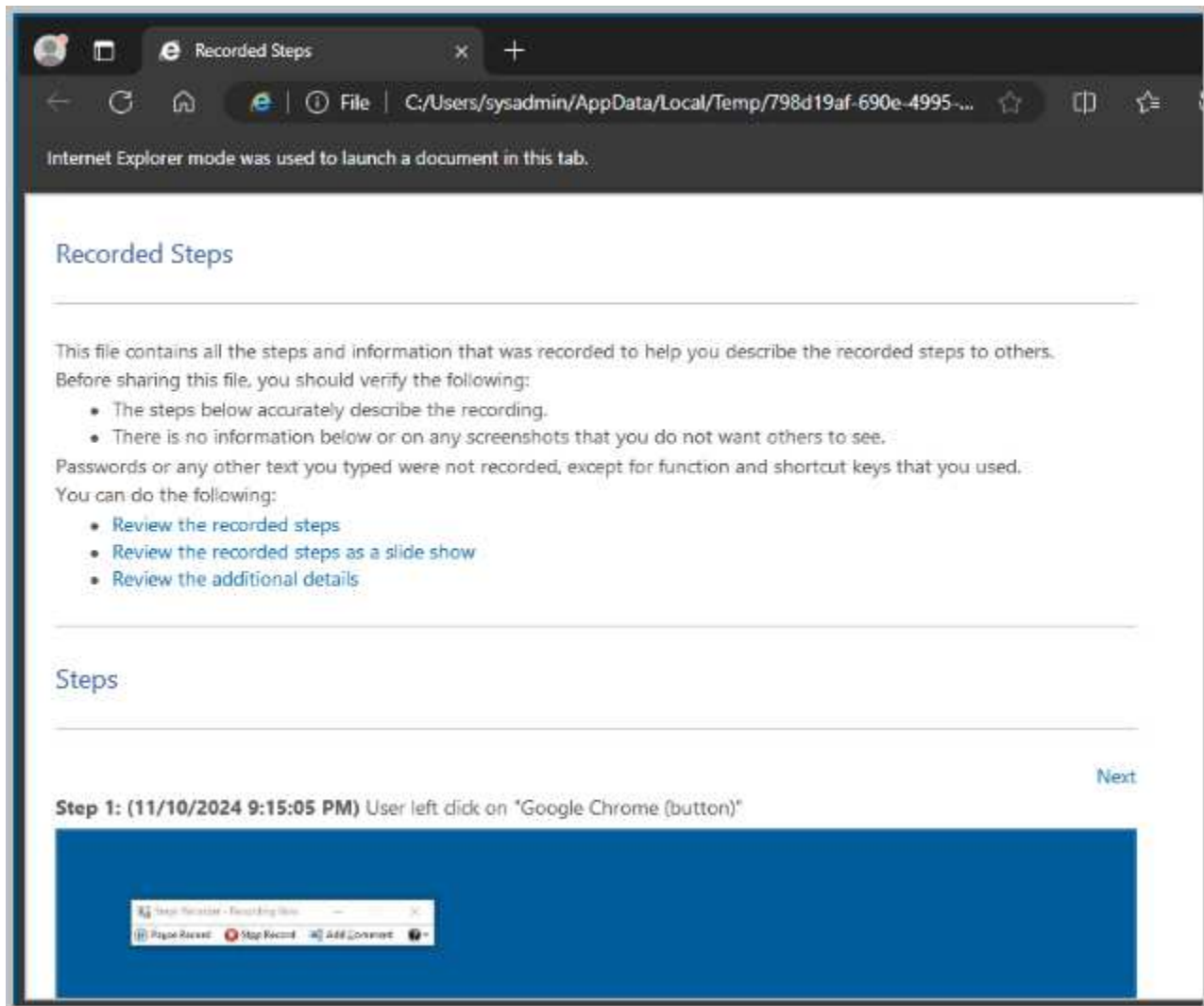
11. Select **Internet Explorer** and click **OK** to open Internet Explorer.



12. Select **Continue** to acknowledge that Internet Explorer has retired and is no longer supported.



13. Review the recorded steps on the **Recorded Steps** tab and notice how much detailed information is available.



Challenge Task:

As a challenge, try to recreate this task in a different version of Windows OS that is made available in this lab environment. This will help show similarities and differences between the versions.

14. The lab is now complete; you may end the reservation.